# glm Â¶

## GeneralizedLinearModelConfig Â¶

```
GeneralizedLinearModelConfig(lambda_search: bool = False, standardize: bool = True, max_after_balance_size: float = 5.0, max_hit_ratio_k: int = 0, compute_p_values: bool = False, balance_classes: bool = False, remove_collinear_columns: bool = False, missing_values_handling: str = 'MeanImputation', solver: str = 'AUTO', non_negative: bool = False, intercept: bool = True, link: str = 'family_default', beta_epsilon: float = 0.0001, theta: float = 1e-10, hglm: bool = False, calc_like: bool = False, max_active_predictors: int | None = None, prior: float | None = None, max_iterations: int | None = None, lambda_min_ratio: float | None = None, objective_epsilon: float | None = None, gradient_epsilon: float | None = None, obj_reg: float | None = None, undersampling: float | None = None, oversampling: float | None = None, seed: int | None = None, ignored_fields: set[str] | None = None, schema: Schema | None = None)
```

Bases: `TrainingConfig`

The H2O Generalized Linear Model configuration object used for model training.

Parameters:

| Name | Type | Description | Default |
|---|---|---|---|
| `lambda_search` Â¶ | `bool` | Use lambda search starting at lambda max, given lambda is then interpreted as lambda min. | `False` |
| `standardize` Â¶ | `bool` | Standardize numeric columns to have zero mean and unit variance. | `True` |
| `max_after_balance_size` Â¶ | `float` | Maximum relative size of the training data after balancing class counts (can be less than 1.0). Requires balance_classes. | `5.0` |
| `max_hit_ratio_k` Â¶ | `int` | Maximum number (top K) of predictions to use for hit ratio computation (for multi-class only, 0 to disable). | `0` |
| `compute_p_values` Â¶ | `bool` | Request p-values computation, p-values work only with IRLSM solver and no regularization. | `False` |
| `balance_classes` Â¶ | `bool` | Balance training data class counts via over/under-sampling (for imbalanced data). | `False` |
| | `bool` | In case of linearly dependent columns, remove some of the dependent columns. | `False` |

| Name | Type | Description | Default |
|---|---|---|---|
| `remove_collinear_columns` ¶ | | | |
| `missing_values_handling` ¶ | `str` | Handling of missing values. Either MeanImputation or Skip. | `'MeanImputation'` |
| `solver` ¶ | `str` | AUTO will set the solver based on given data and the other parameters. IRLSM is fast on on problems with small number of predictors and for lambda-search with L1 penalty, L_BFGS scales better for datasets with many columns. | `'AUTO'` |
| `non_negative` ¶ | `bool` | Restrict coefficients (not intercept) to be non-negative. | `False` |
| `intercept` ¶ | `bool` | Include constant term in the model. | `True` |
| `link` ¶ | `str` | Link. | `'family_default'` |
| `beta_epsilon` ¶ | `float` | Converge if beta changes less (using L-infinity norm) than beta esilon, ONLY applies to IRLSM solver. | `0.0001` |
| `max_active_predictors` ¶ | `int \| None` | Maximum number of active predictors during computation. Use as a stopping criterion to prevent expensive model building with many predictors. Default indicates: If the IRLSM solver is used, the value of max_active_predictors is set to 5000 otherwise it is set to 100000000. | `None` |
| `prior` ¶ | `float \| None` | Prior probability for y==1. To be used only for logistic regression if the data has been sampled and the mean of response does not reflect reality. | `None` |
| `max_iterations` ¶ | `int \| None` | Maximum number of iterations. | `None` |
| `lambda_min_ratio` ¶ | `float \| None` | Minimum lambda used in lambda search, specified as a ratio of lambda_max (the smallest lambda that drives all coefficients to zero). Default indicates: if the number of observations is greater than the number of variables, then lambda_min_ratio is set to 0.0001; If the number of observations is less than the number of variables, then lambda_min_ratio is set to 0.01. | `None` |
| `objective_epsilon` ¶ | `float \| None` | Converge if objective value changes less than this. Default indicates: If lambda_search is set to True the value of objective_epsilon is set to .0001. If the lambda_search is set to False and lambda is equal to zero, the value of objective_epsilon is set to .000001, for any other value of lambda the default value of objective_epsilon is set to .0001. | `None` |

| Name | Type | Description | Default |
| --- | --- | --- | --- |
| `gradient_epsilon` Â¶ | `float` \| `None` | Converge if objective changes less (using L-infinity norm) than this, ONLY applies to L-BFGS solver. Default indicates: If lambda_search is set to False and lambda is equal to zero, the default value of gradient_epsilon is equal to .000001, otherwise the default value is .0001. If lambda_search is set to True, the conditional values above are 1E-8 and 1E-6 respectively. | `None` |
| `obj_reg` Â¶ | `float` \| `None` | Likelihood divider in objective value computation, default is 1/nobs. | `None` |
| `theta` Â¶ | `float` | Theta. | `1e-10` |
| `hglm` Â¶ | `bool` | If set to true, will return HGLM model. Otherwise, normal GLM model will be returned. | `False` |
| `calc_like` Â¶ | `bool` | If true, will return likelihood function value for HGLM. | `False` |
| `undersampling` Â¶ | `float` | The value for undersampling. Only this or oversampling can be used. | `None` |
| `oversampling` Â¶ | `float` | The value for oversampling. Only this or undersampling can be used. | `None` |
| `seed` Â¶ | `int` | The seed value. | `None` |
| `ignored_fields` Â¶ | `set of str` | Set of names corresponding to the fields to be ignored when training a model. | `None` |
| `schema` Â¶ | `Schema` | Pass a schema to enable conversion between descs and internal names. By not passing a schema, ds-api will assume field names are references to internal Pulse names, not the ones in the UI. | `None` |

Examples:

```
>>> model_config = GeneralizedLinearModelConfig()
>>> model_config.remove_collinear_columns
>>> model_config.remove_collinear_columns = True # sets it to True
>>> model = model_project.train(
...     "model_name",
...     datasource,
...     "target_field",
...     model_config
... )
>>> model.waitUntilModelTrained()
>>> config_from_model = model.get_config()
```

### balance_classes `property` `writable` Â¶

```
balance_classes
```

### beta_epsilon `property` `writable` Â¶

`beta_epsilon`

### calc_like `property` `writable` Â¶

`calc_like`

### compute_p_values `property` `writable` Â¶

`compute_p_values`

### gradient_epsilon `property` `writable` Â¶

`gradient_epsilon`

### hglm `property` `writable` Â¶

`hglm`

### ignored_fields `property` `writable` Â¶

`ignored_fields`

### intercept `property` `writable` Â¶

`intercept`

### lambda_min_ratio `property` `writable` Â¶

`lambda_min_ratio`

### lambda_search `property` `writable` Â¶

`lambda_search`

### link `property` `writable` Â¶

`link`

### max_active_predictors `property` `writable` Â¶

`max_active_predictors`

### max_after_balance_size `property` `writable` Â¶

`max_after_balance_size`

### max_hit_ratio_k `property` `writable` Â¶

`max_hit_ratio_k`

### max_iterations `property` `writable` Â¶

`max_iterations`

### missing_values_handling `property` `writable` Â¶

`missing_values_handling`

### non_negative `property` `writable` Â¶

```
non_negative
```

### obj_req `property` `writable` Â¶

```
obj_req
```

### objective_epsilon `property` `writable` Â¶

```
objective_epsilon
```

### oversampling `property` `writable` Â¶

```
oversampling
```

### prior `property` `writable` Â¶

```
prior
```

### remove_collinear_columns `property` `writable` Â¶

```
remove_collinear_columns
```

### seed `property` `writable` Â¶

```
seed
```

### solver `property` `writable` Â¶

```
solver
```

### standardize `property` `writable` Â¶

```
standardize
```

### subclasses `class-attribute` `instance-attribute` Â¶

```
subclasses: List[JavaWrapper] = []
```

### theta `property` `writable` Â¶

```
theta
```

### undersampling `property` `writable` Â¶

```
undersampling
```

### auto_cast `staticmethod` Â¶

```
auto_cast(config: JavaWrapper) -> GeneralizedLinearModelConfig
```

Creates a GeneralizedLinearModelConfig from the java object.

Parameters:

| Name | Type | Description | Default |
|---|---|---|---|
| `config` Â¶ | `JavaWrapper` | Java configuration object. | *required* |

Returns:

| Type | Description |
| --- | --- |
| `GeneralizedLinearModelConfig` | Configuration object according to its python representation. |